

OSLO GARDERMOEN, Norway

WMO: 013840

Lat: 60.206N

Lon: 11.080E

Elev: 208

StdP: 98.86

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.8	-16.2	-20.8	0.6	-18.5	-18.2	0.8	-15.8	10.8	2.5	9.4	1.6	1.1	20	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	26.2	16.5	24.4	15.9	22.5	15.1	18.1	23.0	17.1	21.9	16.2	20.7	3.5	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	11.8	19.0	15.3	11.2	18.5	14.4	10.5	17.7	51.7	23.0	48.7	21.7	46.2	20.9	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	7.9	6.8	DB	-21.5	28.2	3.0	2.0	-23.7	29.6	-25.5	30.8	-27.2	32.0	-29.4	33.4	(4)
(5)			WB	-21.7	19.3	3.0	1.2	-23.9	20.2	-25.6	20.9	-27.3	21.6	-29.5	22.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-5.0	-4.5	-0.6	4.6	9.9	14.0	16.7	15.4	11.1	5.3	0.3	-3.7	(6)	
	DBStd	8.67	5.56	5.45	4.41	3.32	3.76	2.79	2.75	2.68	2.97	3.58	4.19	5.57	(7)	
	HDD10.0	2303	464	406	329	164	48	2	0	0	23	149	291	426	(8)	
	HDD18.3	4770	723	639	588	411	262	135	67	99	218	404	541	684	(9)	
	CDD10.0	602	0	0	0	3	45	121	208	167	55	4	0	0	(10)	
	CDD18.3	28	0	0	0	0	1	3	17	7	0	0	0	0	(11)	
	CDH23.3	277	0	0	0	0	14	48	157	57	0	0	0	0	(12)	
	CDH26.7	35	0	0	0	0	1	6	26	3	0	0	0	0	(13)	
(14) Wind	WSAvg	2.9	2.5	2.7	2.9	3.1	3.3	3.3	2.9	2.7	2.9	2.7	2.7	2.5	(14)	
(15) Precipitation	PrecAvg	866	62	47	49	49	66	78	82	96	88	97	90	64	(15)	
	PrecMax	1121	157	146	135	102	156	176	209	220	211	216	299	154	(16)	
	PrecMin	582	4	5	2	0	1	10	7	5	23	10	15	12	(17)	
	PrecStd	141	36	30	31	28	35	40	41	52	44	55	53	30	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.2	7.1	13.5	18.7	25.3	27.2	29.2	27.0	21.4	15.0	11.0	8.0	(19)	
		MCWB	4.1	3.2	7.2	10.3	15.1	16.2	17.6	16.6	15.2	11.1	8.9	6.7	(20)	
	2%	DB	4.3	5.0	9.7	15.9	22.1	24.7	27.0	25.0	19.2	13.1	9.1	6.2	(21)	
		MCWB	3.0	2.5	4.9	8.9	13.1	15.0	17.1	16.5	13.4	10.6	7.9	5.2	(22)	
	5%	DB	3.1	3.8	7.0	13.2	19.8	22.1	25.0	23.0	17.7	11.9	7.6	4.8	(23)	
		MCWB	2.1	2.2	3.1	7.4	12.1	13.7	16.6	15.8	12.9	9.8	6.6	3.8	(24)	
	10%	DB	2.0	2.5	5.0	11.0	17.2	20.0	23.0	20.9	16.0	10.8	6.1	3.2	(25)	
		MCWB	1.1	1.4	2.1	6.2	10.7	12.9	15.7	14.9	12.4	9.1	5.2	2.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	4.5	7.5	11.6	16.2	17.4	19.5	18.7	16.6	12.7	9.7	7.1	(27)	
		MCDB	5.3	5.6	12.8	16.7	22.2	23.1	24.9	23.0	19.0	14.0	10.3	7.5	(28)	
	2%	WB	3.4	3.2	5.3	9.6	14.2	16.1	18.5	17.7	15.1	11.5	8.1	5.4	(29)	
		MCDB	3.9	4.1	8.6	14.3	20.0	21.9	24.1	21.8	17.7	12.6	8.7	6.0	(30)	
	5%	WB	2.2	2.4	3.9	8.2	12.9	14.9	17.5	16.8	14.0	10.5	6.7	3.9	(31)	
		MCDB	2.8	3.3	6.0	12.0	17.4	20.3	22.9	20.9	16.3	11.5	7.3	4.4	(32)	
	10%	WB	1.1	1.4	2.6	6.9	11.6	13.9	16.6	16.0	13.0	9.4	5.2	2.4	(33)	
		MCDB	1.7	2.3	4.5	10.1	16.5	18.7	21.5	19.8	15.2	10.4	5.9	3.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.5	8.1	9.0	9.7	9.2	9.2	8.8	8.0	6.2	4.6	5.2	(35)	
		MCDBR	5.6	6.5	10.4	13.4	14.1	13.2	13.1	12.0	10.2	7.1	6.4	5.8	(36)	
	5% WB	MCWBR	5.0	5.1	6.7	7.6	6.8	6.0	5.4	5.3	5.8	5.2	5.8	5.5	(37)	
		MCWBR	5.1	5.6	9.2	11.5	11.9	10.8	10.9	9.9	8.2	5.9	5.8	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.252	0.295	0.322	0.365	0.363	0.353	0.373	0.365	0.333	0.319	0.272	0.224	(40)	
		taud	2.195	2.274	2.318	2.315	2.403	2.452	2.413	2.430	2.511	2.457	2.277	2.102	(41)	
	Ebn at Noon		528	706	806	831	864	879	848	826	799	681	501	400	(42)	
		Edh at Noon	40	66	87	105	104	101	103	94	73	56	37	27	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.28	0.90	2.37	3.76	4.83	5.52	4.94	3.88	2.57	1.14	0.37	0.18	(44)	
	RadStd		0.05	0.17	0.25	0.50	0.50	0.43	0.65	0.33	0.30	0.22	0.10	0.02	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	+2.03	+1.93	N/A	N/A	N/A	-177	N/A	N/A	N/A	
(47) Regional (56 neighbors)		N/A	N/A	+1.55	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page